

Operation Manual

pH value measuring system

ORKY-pH/T



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This Operation Manual was prepared whilst taking into account the ATV [German Effluent Treatment Organization] Instruction Sheet No. M 269 "Requirements on on-line process analysis instruments for N and P", as well as the ATV Instruction Sheet No. M 256 "Use of operational measuring devices on sewage treatment plants - preliminary sheet".

1 General

1.1 Introduction

Figure 1 shows the construction of the measurement system for pH. The system consist of the transducer ORKY and the pH probe with cable.

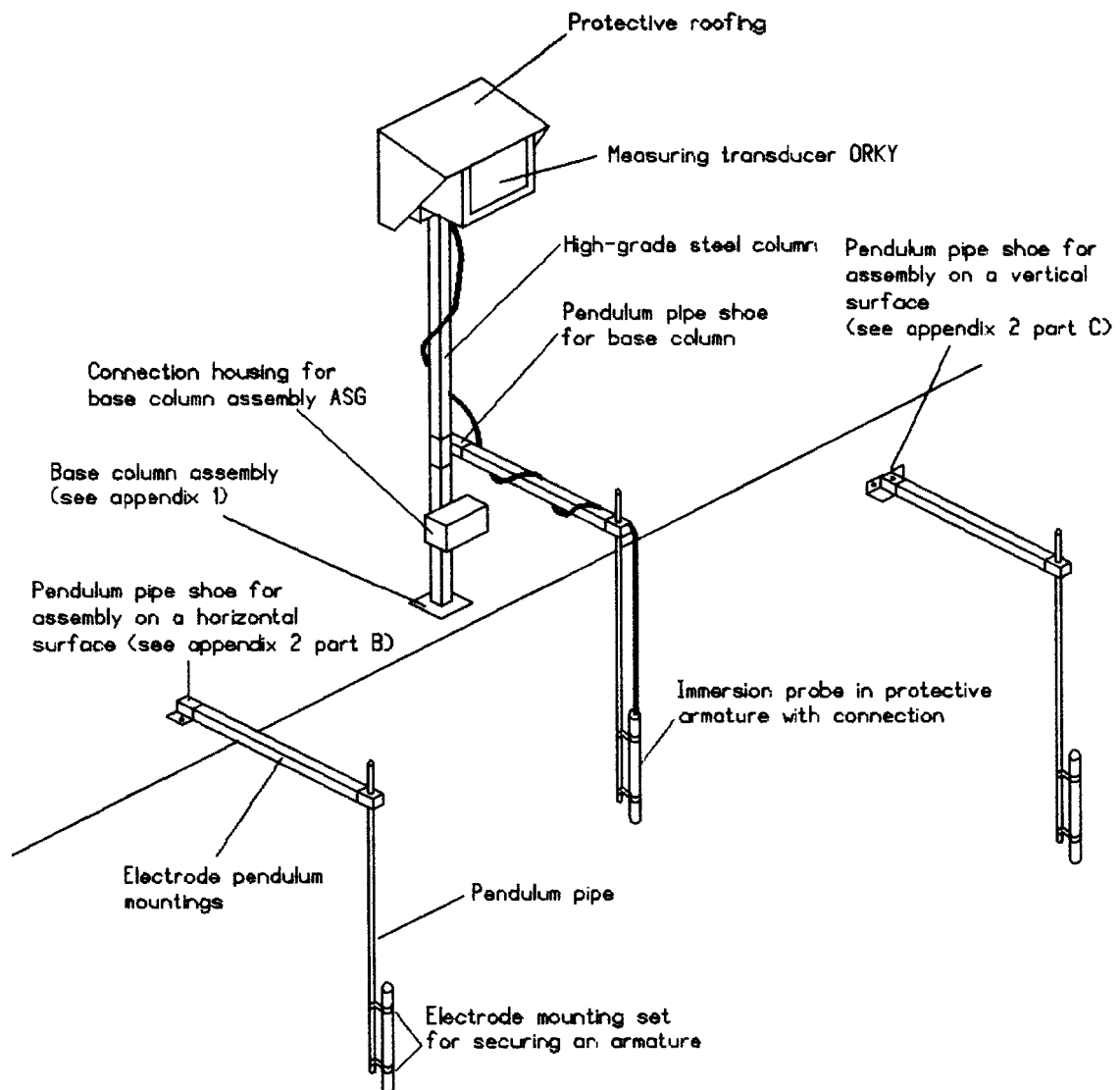


Figure 1: Measurement transducer ORKY with pH probe

Table 1 contains the technical data.

Table1: Technical data

Type		ORKY pH/T
Measuring range	pH	0-14 or 2-12, selectable
	Temperature	0° C – 100° C, selectable
Accuracy ORKY	Display	± 0,4 % of full scale
	Signal Output	± 0,4 % of full scale
Accuracy Probes	pH	+ - 0,2 pH
	Temperature	+ - 0,5% F.S.
Response time	pH	< 5 secs.
	Temperature	< 5 secs.
Analogous inputs		8, galvanically isolated
Relais output		4, optional 12, change-over contact, potential free
Current outputs	pH	0/4-20 mA, max. load 500 Ω
	Temperature	0/4-20 mA, max. load 500 Ω
Interface		RS 232
Ambient temperature		+2° ... +45 °C
Protection class		IP 54
Power supply		230 V AC, +10/-15%, 50 Hz, 24 V DC
Max. power consumption		15 VA
Cross section for connection		1,5 mm ²
Weight of ORKY		2 kg

1.2 Safety instructions

1.2.1 Indented use

The ORKY transducer from GIMAT GmbH Umweltmesstechnik is an transducer for analog signal input and with analog and digital signal output.

Any other use or a use going beyond this counts is improper. The operator may operate and maintain the instrument only taking into account all the advice for use in the present operation manual, and after instruction has been given by service personnel authorized by the manufacturer.

The operator is obliged to comply absolutely with the prescribed maintenance operations and the associated exchange of components subject to wear.

Applications going beyond the envisaged use are inadmissible. GIMAT GmbH Umweltmesstechnik accept no liability for damages which are caused by improper use of the instrument.

Proper use also includes:

- the observance of all the instructions in the operation manual,

- the observance of maintenance operations,
- carrying out inspections by service personnel,
- the observance of the general and specific safety advice in this operation manual, as well as the relevant regulations for preventing accidents.

1.2.2 Observe instructions in the operation manual

Safe handling and satisfactory operation of the ORKY transducer are dependent on being acquainted with all the fundamental safety regulations.

The operation manual, in particular the safety instructions, are to be observed by all those persons who operate the instrument and produce calibration solutions and/or reagents.

In addition, it is necessary to observe the rules and regulations for accident prevention which apply at the location of use.

The operator checks the safety-aware work of the personnel at regular intervals.

1.2.3 Obligations of the personnel

Persons who are charged with working on the ORKY transducer commit themselves, before starting work:

- to observe the regulations about health and safety at work and accident prevention,
- to read the safety Chapter and the warning instructions in this operation manual and to observe these continuously during installation, commissioning and operation.

1.2.4 Hazards in dealing with the instrument and chemicals

The ORKY transducer instruments have been constructed in accordance with recognized safety rules. The instrument is to be used only

- for the proper application
- in a satisfactory safety state.

In the case of improper use, hazards may arise to the health of the user or third parties (severe damage to the skin or to the eyes) and/or impairments to the instrument, or to other material assets.

GIMAT mbH Umweltmesstechnik accepts no liability for damages which arise from improper dealings with the instrument or chemicals.

Faults which can impair safety should be eliminated without delay.

1.2.5 Warranty and liability

The "general business conditions" of GIMAT GmbH Umweltmesstechnik apply in principle. These are available to the operator at the latest from the time of concluding the contract.

Guarantee and liability claims are ruled out if they can be traced back to one or more of the following causes:

- improper use of the instrument,
- improper installation, commissioning, operation and maintenance of the instrument,
- operation of the instrument in the event of defective and/or non-functioning safety and protective equipment,
- non-observance of the instructions in the operation manual relating to installation, commissioning, operation and maintenance of the instrument,
- unauthorized constructional changes to the instrument,
- deficient monitoring of instrument parts which are subject to wear,
- repairs carried out improperly,
- catastrophes resulting from the action of foreign bodies and Acts of God.

1.2.6 Informal safety measures

In addition to the operation manual, the generally applicable and the local regulations for accident prevention and for environmental protection are to be provided and observed.

Check the following several times a week:

- all visible instrument components for detectable external damage,
- all electric feed lines for loose connections, scorch points and other damage,
- all mechanical connection points for leaks.

1.2.7 Hazards from electrical power

The electrical connection of the instrument should be checked regularly. Loose connections should be corrected at once and scorched cables should be exchanged by specialist personnel.

An electrical hazard is presented by direct or indirect electric contact. This is to be borne in mind, particularly during the installation of the electrical connection.

This may be carried out only by trained electrical technical personnel.

1.2.8 Maintenance and overhaul, rectification of faults

It is necessary to carry out the maintenance operations described in Chapter 6 of this operation manual, so that low-fault operation is ensured in the long term.

If faults occur during operation with the ORKY transducer, then information should be sought in this operation manual, Chapter 7, about causes of the fault and the possibility for its rectification.

If a fault occurs in your instrument, and it cannot be rectified using the information in this operation manual, then please contact the Service Department of GIMAT mbH Umweltmesstechnik (address: see below)

No changes, additions or rebuilding of the ORKY transducer may be undertaken without authorization from GIMAT mbH Umweltmesstechnik.

Instrument parts in an unsatisfactory condition should be exchanged immediately.

Only original replacement or original wearing parts may be used. In the case of parts procured elsewhere, it is not guaranteed that they have been constructed and prepared to comply with the demands and safety.

For guarantee claims, repair and spare-part service, please contact:

GIMAT GmbH Umweltmesstechnik
Obermuehlstrasse 70
82398 Polling
Germany
Tel. +49-881-628-0
Fax. +49-881-628-15
e-mail: gimat2000@aol.com
Internet: www.gimat.de

2 Functional description

The functional principle of the measurement transducer ORKY is shown in Figure 2.

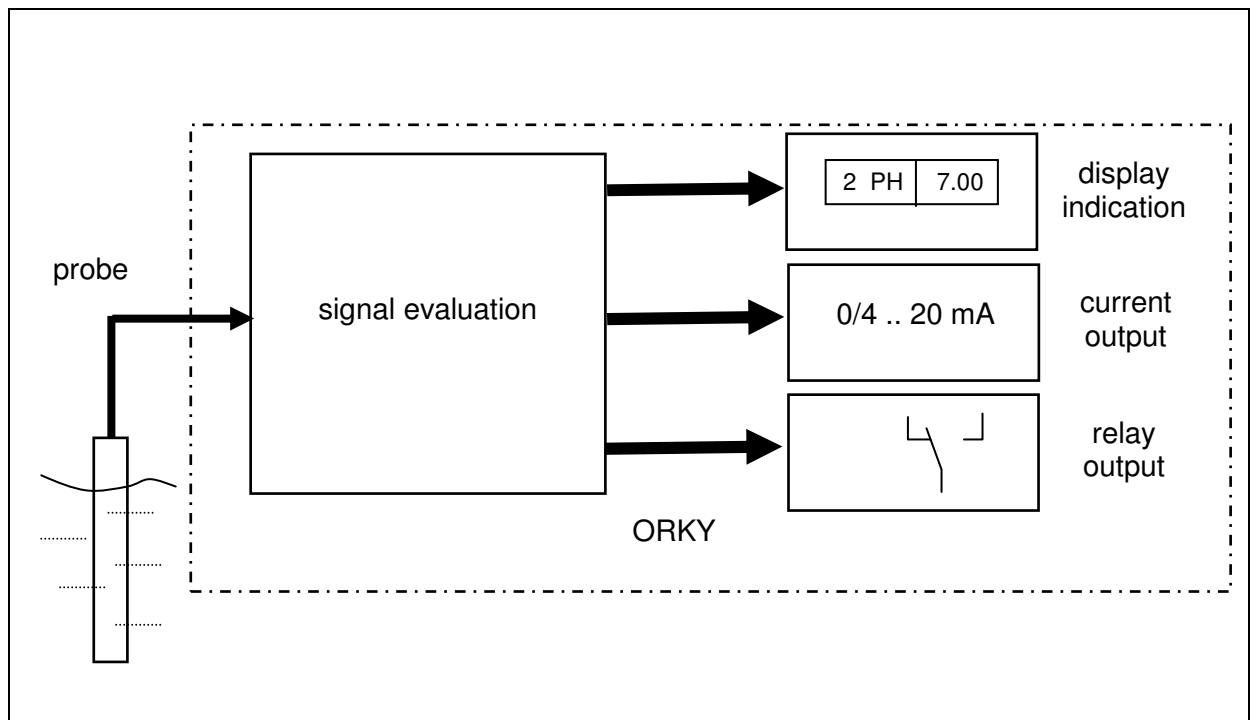


Figure 2: Functional principle of the measurement transducer ORKY

3 Electrical Connection

The electrical connection must only be carried out by suitably qualified personnel and in accordance with the VDE guidelines.

It is absolutely essential that the assembly instructions be observed during connection and pre-assembly.

The connection is made to the terminal connection strip as per the terminal plan. Access to the terminal strip is possible after opening the front panel of the housing. Before opening the power supply must be switched off!

Cable leads not in use must be sealed off, in order to prevent damage from penetrating moisture.

Following completion of the installation, the main power supply and the ORKY-pH/T should be switched on. For operation, the power supply must be ensured at all times.

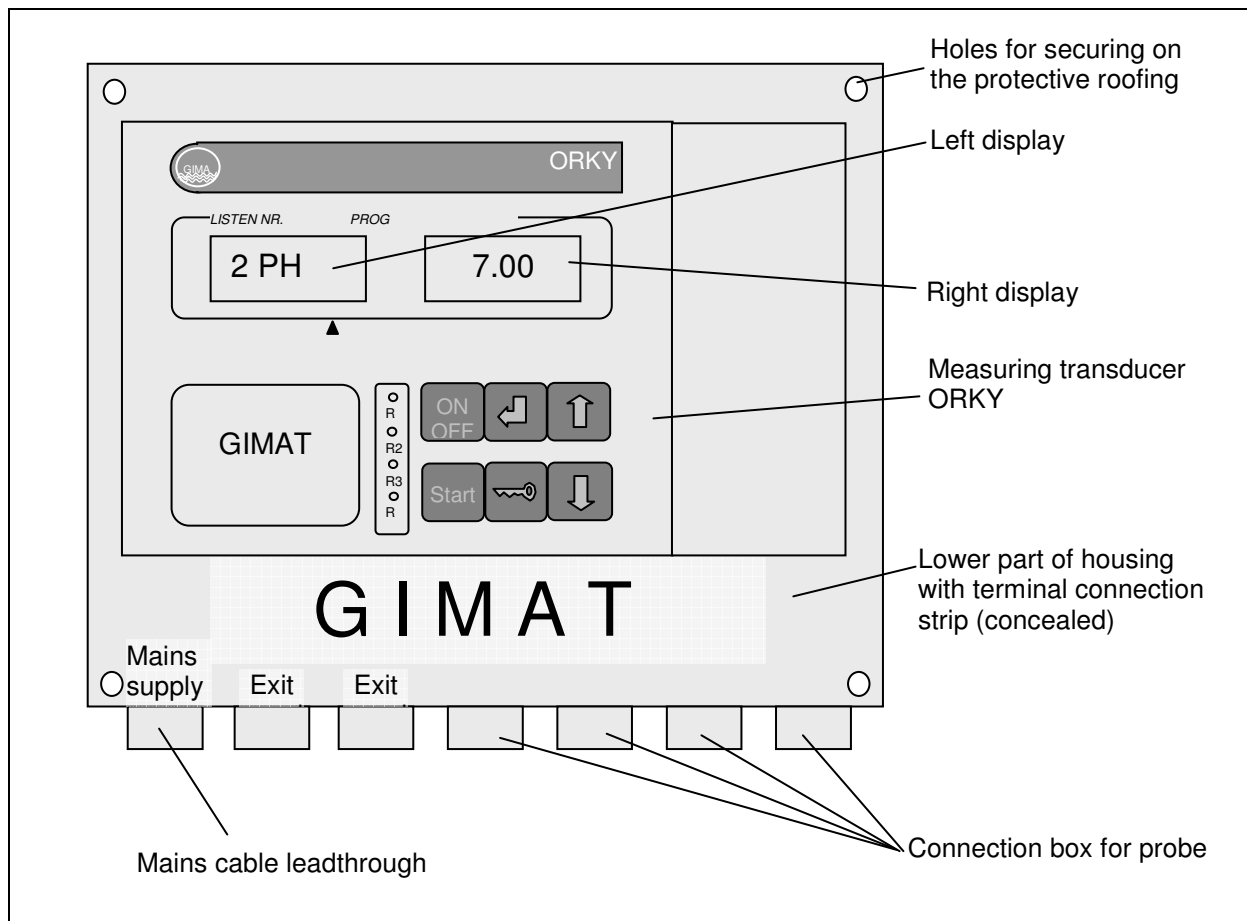


Figure 3: Electrical connection

4 Commissioning

4.1 Measuring transducer ORKY

The measuring transducer ORKY is the central evaluation unit of the pH measuring system. All functions of the device, as well as the output of the measured values and signals are controlled by the ORKY. Likewise, the calibration values are entered into the measuring transducer and stored.

4.1.1 Meaning of the buttons on the ORKY

ON/OFF	Switching the ORKY on/off
↵ [ENTER]	Calling up a report point, confirming the flashing display value
↑	Increasing the flashing display value
↓	Reducing the flashing display value
🔑	Preparing the entering of parameters or calibration

4.1.2 Calling up report points

- Press ↵
- Select the flashing value with ↑ or ↓ and thereafter confirm with ↵

Using 🔑 the input parameters as per Table 4 can be opened and the settings altered. When doing so, the respective flashing value can be altered using the ↑ and ↓ and thereafter confirmed with ↵.

After a selection of the report point, the corresponding indication appears in the display.

4.2 Commissioning diagram

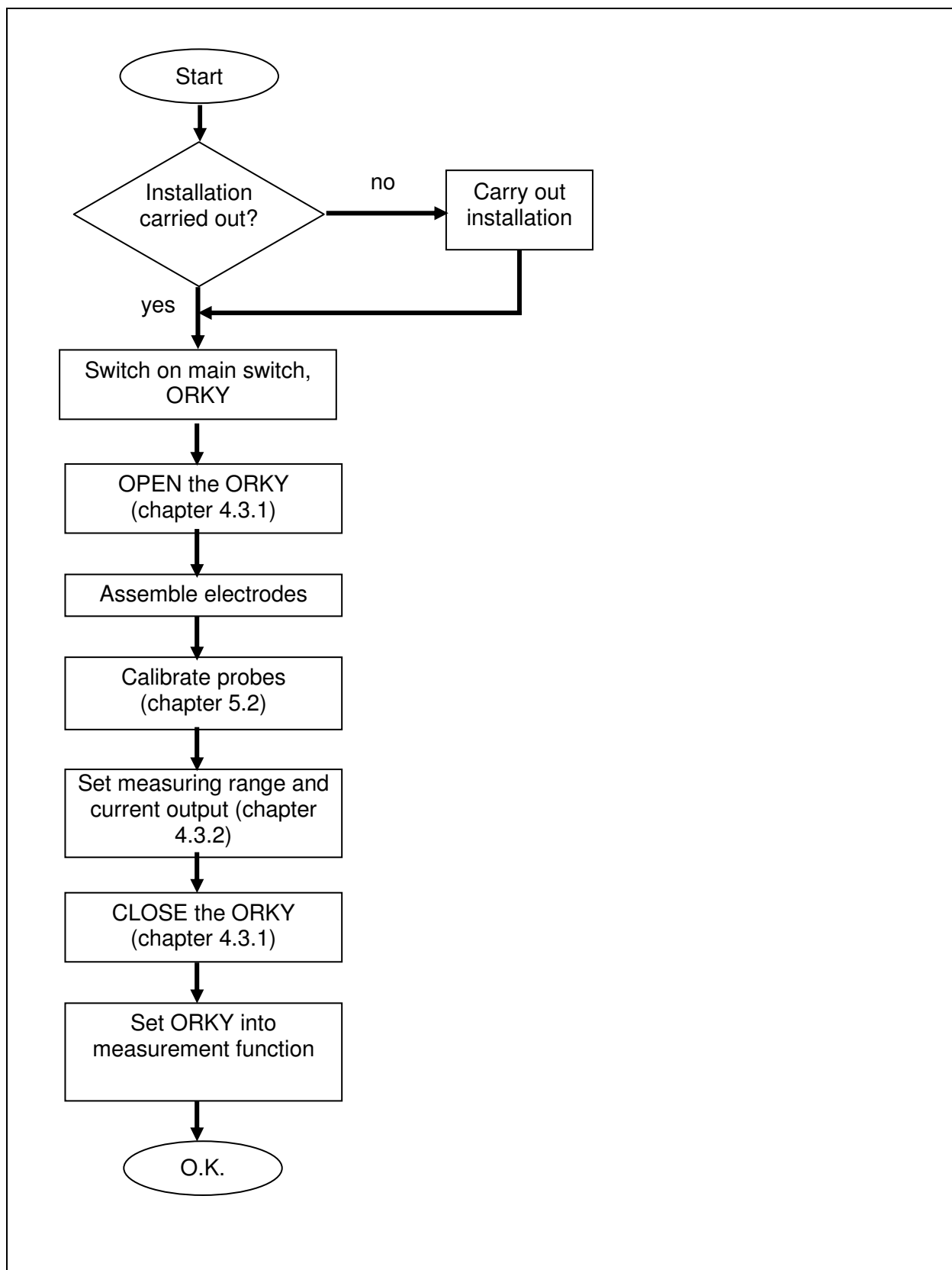


Figure 4: Diagram for commissioning

4.3 Setting the input and output data

During commissioning, the input and output data of Table2 should be checked and set.

Table2: Input and output data

Report point	Meaning	see chapter ...
00	Code, open or close ORKY	4.3.1
11	Measuring value pH;	4.3.2
20	Current output Scale of current output	4.3.2
40	Showing/setting time and date	4.4
43	Activation/setting of scrolling display	-
54	Activation of automatic temperature compensation	5.2.2
58	Showing/setting of reference temperature	4.3.3
60	Carrying out calibration	5.2.3
89	Service- and error messages	8

4.3.1 Opening and closing the ORKY

The ORKY can be opened and closed under report point **00**. The following procedure is recommended:

Table3: Opening and closing the ORKY

	Indication left display	Indication right display	Status of device
1. Request the status of the device by selecting report point 00 as per chapter 4.1.2			
Either	or	CLOs	„closed“
or	or	OPEn	„open“
- Confirm with  when the device has the desired status. - Continue working as per point 2 or 3 the status of the device is to be altered.			
Action	Indication left display	Indication right display	
2. Altering the status of the device from closed to open:			
- Press 	CODE	5555	
- Enter the correct code. The respective flashing number can be altered by pressing the  or  .			
	or	OPEN	
	00		
- You are now again in the report point level!			
3. Altering the status of the device from open to closed			
- Press 	or	<u>OPEN</u>	
- The flashing „ <u>OPEN</u> “ can be altered to „CLOS“ by pressing  or  .			
	or	CLOs	
	00		
 You are now again in the report point level!			

4.3.2 Setting the measuring ranges and the current output

The current output range and the measuring range can be set under report point **21**. The following procedure is recommended:

- Open the ORKY as per chapter 4.3.1.
- Select report point 20 and press .

Table 4: Setting of current outputs

Indication left display	Indication right display	Alternation of the ...
bEr	Either „0..20“ or „4..20“ or „20..0“ or „20..4“ or „OFF“	Current exit by selection of $\uparrow$ or $\downarrow$ and confirmation of each number with $\rightarrow$
o DO	A four figure number consisting of 4 numbers	Measuring range upper limit by selection of $\uparrow$ or $\downarrow$ and confirmation of each number with $\rightarrow$
u DO	A four figure number consisting of 4 numbers	Measuring range lower limit by selection of $\uparrow$ or $\downarrow$ and confirmation of each number with $\rightarrow$

- Close the ORKY as per chapter 4.3.1

The value set in this manner for the measuring range upper limit „o PH“ corresponds to the current exit 20 mA. The value set in this manner for the measuring range lower limit „u PH“ corresponds to the current exit 0 or 4 mA.

4.3.3 Setting of reference temperature

For the temperature compensation of the pH measurement the temperature of the water has to be set. Changes can be done according Table 5.

Table 5: Setting of reference temperature

- Select report point 54 with $\rightarrow$, $\uparrow$ or $\downarrow$			
Indication left display	Indication right display	Wish	Necessary action
°Cr	e. g. 25.0	Correct temperature	- press  - if a code is requested, set code number - select correct reference temperature (°C) with $\rightarrow$, $\uparrow$ or $\downarrow$ - confirm with $\rightarrow$
- Return to report point level with $\rightarrow$			

4.4 Select time

An internal clock is integrated in ORKY. You can set the present time and date according following procedure.

Table 6: Setting of clock

- Select report point 40 with buttons $\leftarrow$, $\uparrow$ or $\downarrow$			
Indication left display	Indication right display	Wish	Necessary action
Uhr	Time (hours: minutes) e. g. 7:35	Different time than indicated	- press  - If code is requested, set code number - Set new time with $\leftarrow$, $\uparrow$ or $\downarrow$ and confirm with $\leftarrow$
- Return to report point level with $\leftarrow$ or - Press $\uparrow$ for selecting seconds			
SEC	Seconds, e. g. 25	Different time than indicated	- press  - If code is requested, set code number - Set seconds with $\leftarrow$, $\uparrow$ or $\downarrow$ and confirm with $\leftarrow$
- Return to report point with $\leftarrow$ or - Press $\uparrow$ for selecting day and month			
dAtu	Date e. g. 26.02	Different day and month than indicated	- press  - If code is requested, set code number - Set day and month with $\leftarrow$, $\uparrow$ or $\downarrow$ and confirm with $\leftarrow$
- Return to report point level with $\leftarrow$ or - Press $\uparrow$ for selecting year:			
JAhr	Year, e. g. 2002	Different year than indicated	- press  - If code is requested, set code number - Set year with $\leftarrow$, $\uparrow$ or $\downarrow$ and confirm with $\leftarrow$
Press $\uparrow$ for indication of time or $\leftarrow$ for returning to report point level			

5 Operation

5.1 Display of the measured value

Select report point 13 (see Table 4 and chapter 4.1)

The left display shows the number of the input channel and a symbol for the measured parameter. The measured value is shown in the right display.

Note The ORKY should always be closed “CLOs” be during operation (see chapter 4.3.1)

The ORKY must be set to one of the report point 11 after each setting of the parameter or calibration.

5.2 Calibration of the pH probe

For a correct measurement of the pH value the pH electrode had to be calibrated according following procedure:

- Switching on or off the automatic temperate compensation respectively
- Selecting the reference temperature
- Checking the settings and correcting the settings if necessary
- Calibrating the probe

In case of the temperature sensor a calibration is not necessary.

5.2.1 Selecting the calibration solutions

Two pH calibration solutions are needed. One of the calibration solutions should have a pH value of approximately 7, e.g. GIMAT pH calibration solution 6.88. The second calibration solution should have a pH value of about 4 (e. g GIMAT pH calibration solution 4.00) or 10 (e.g GIMAT pH calibration solution 9.22) according to the measurement range.

5.2.2 Setting of the parameter

Selecting report point 50 the setting of the pH calibration points can be checked and be corrected if necessary.

The automatic temperature compensation can be switched off or on „8 °C“ in report point 54. If the temperature sensor of the ORKY is used for the calibration of the pH electrode switch temperature sensor on („8 °C“), otherwise switch temperature sensor off.

Table 7 shows the procedure for setting the pH calibration points.

Table 7: Setting of the pH calibration points

Indication left display	Indication right Display	wish	necessary action
1 PH	actual calibration point 1, e. g. 6.88	Alter the value	- press  - set pH-value of calibration solution 1 at reference temperature by pressing $\uparrow$ or $\downarrow$ and $\rightarrow$ - Confirm by pressing $\rightarrow$
- Select the next pH calibration point by pressing $\uparrow$			
2 PH	actual calibration point 2, e. g. 9.22	Alter the value	- press  - set pH-value of calibration solution 2 at reference temperature by pressing the $\uparrow$ or $\downarrow$ and $\rightarrow$ - Confirm by pressing $\rightarrow$
- You are now again in the report point level (report point 50)			
- Select the report point 54 by pressing $\uparrow$ or $\downarrow$ and $\rightarrow$			
Corr	„OFF“ or „8 °C“	Alter the value	- press  - change setting by pressing the $\uparrow$ or $\downarrow$ and $\rightarrow$ - Confirm by pressing $\rightarrow$
- You are now again in the report point level (report point 54)			

5.2.3 Carrying out the pH calibration

In the first step the pH probe had to be taken out of the pipe and cleaned carefully with water.

Calibration is carried out under report point 60. The following procedure is recommended (see table 8):

Table 8: Carrying out calibration of the pH electrode

Indication left display	Indication right Display	necessary action
°Cr	Value for the temperature	- Set temperature of the calibration solution be pressing $\uparrow$ or $\downarrow$ and $\rightarrow$ - Confirm by pressing $\rightarrow$
1 PH	Calibration point 1	- Immerse pH-electrode in calibration solution 1 - Immerse temperature sensor in pH calibration solution 1, if automatic temperature sensor is switched on („8 °C“) - Confirm by pressing $\rightarrow$
CAL 1	calibration value	- Wait until the value stand still - Confirm by pressing $\rightarrow$
2 PH	Calibration point 2	- Immerse pH-electrode in calibration solution 1 - Immerse temperature sensor in pH calibration solution 2, if automatic temperature sensor is switched on („8 °C“) - Confirm by pressing $\rightarrow$
CAL 2	calibration value	- Wait until the value stand still - Confirm by pressing $\rightarrow$
- In case of a successful calibration you are back again in the report point level (report point 60). - Close ORKY under report point 00. - Select report point 11 to display the measured pH value.		

6 Maintenance

An overview about regular maintenance is given in Table 9.

Table 9: Required regular maintenance

Interval	Task	see chapter
On demand ca. every 1 - 6 months	Checking the function	6.1
	Rinsing the probes (depending on the content of suspended solids etc.)	6.2
	Calibration of pH electrode	5.2
On demand; approximately every 1-2 years	Replacement of pH probe	

6.1 Supervising of functions

In order to prevent problems and error messages a regular supervising is recommended, especially that

- sensor tips must be always covered by water
- all mechanical and electrical connections tight enough in order to prevent leakages, intermittent contacts etc.

6.2 Dismantling of probes

The pH electrode is made out of glass. Handle with care!

6.3 Cleaning of the probes

The probes should be cleaning with drinking water.

Attention:	Do not use cleaning powder! Do not pumice sensor tips!
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7 Replacement parts

Replacement parts are summarized in Table 10. These can be obtained from GIMAT mbH Water Monitors.

Table 10: Replacement parts

Description	Abbreviated designation	Quantity	GIMAT order No.
pH-measuring chain, 1 diaphragma, with plug-in head	pH 722 Z-1-Gel/Stk.	1 pc.	9910 0000
pH- measuring chain, 3 diaphragma, with plug-in head	pH 722 Z-3-Gel/Stk.	1 pc.	9912 0000
Buffer solution pH 4.00	pH-P 4.00	1000 ml	9914 0000
Buffer solution pH 6.88	pH-P 6.88	1000 ml	9915 0000
Buffer solution pH 9.22	pH-P 9.22	1000 ml	9916 0000
Cleaning solution for pH probe	pH-RL	1000 ml	9917 0000
Cable for pH-electrode, with connector	pH-K-Stk	1 pc.	9909 0xv0
Temperature probe, d=12 mm, V2A, length 180 mm	Pt 100/18	1 pc.	9940 0xv0

Comments about order No.: x=1: without plug V=1.. 6: standard cable length
x=2: DIN-plugV=9: special cable length on request
x=3: water proofed plug

Orders can be placed by telephone on tel.: +49-881-6280 or by facsimile on +49-881-62815 or e-mail on gimat2000@aol.com.

8 Notification of disturbances, location and correction of faults

The ORKY issues a notification of disturbances on the display of the ORKY, whenever measuring is disturbed. Figure 5 shows the possible disturbance notifications and symptoms as well as instructions for locating and correcting faults.

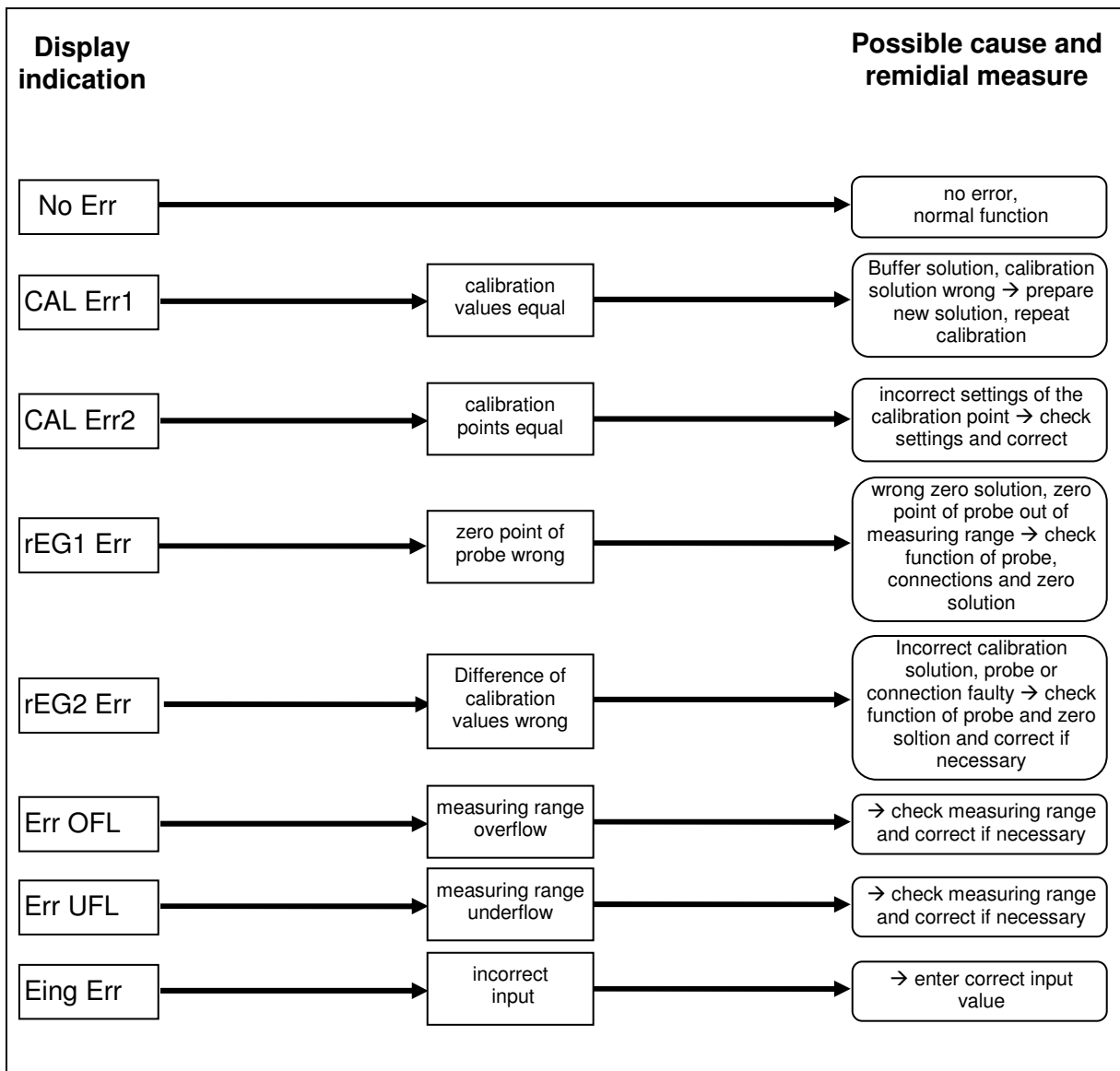
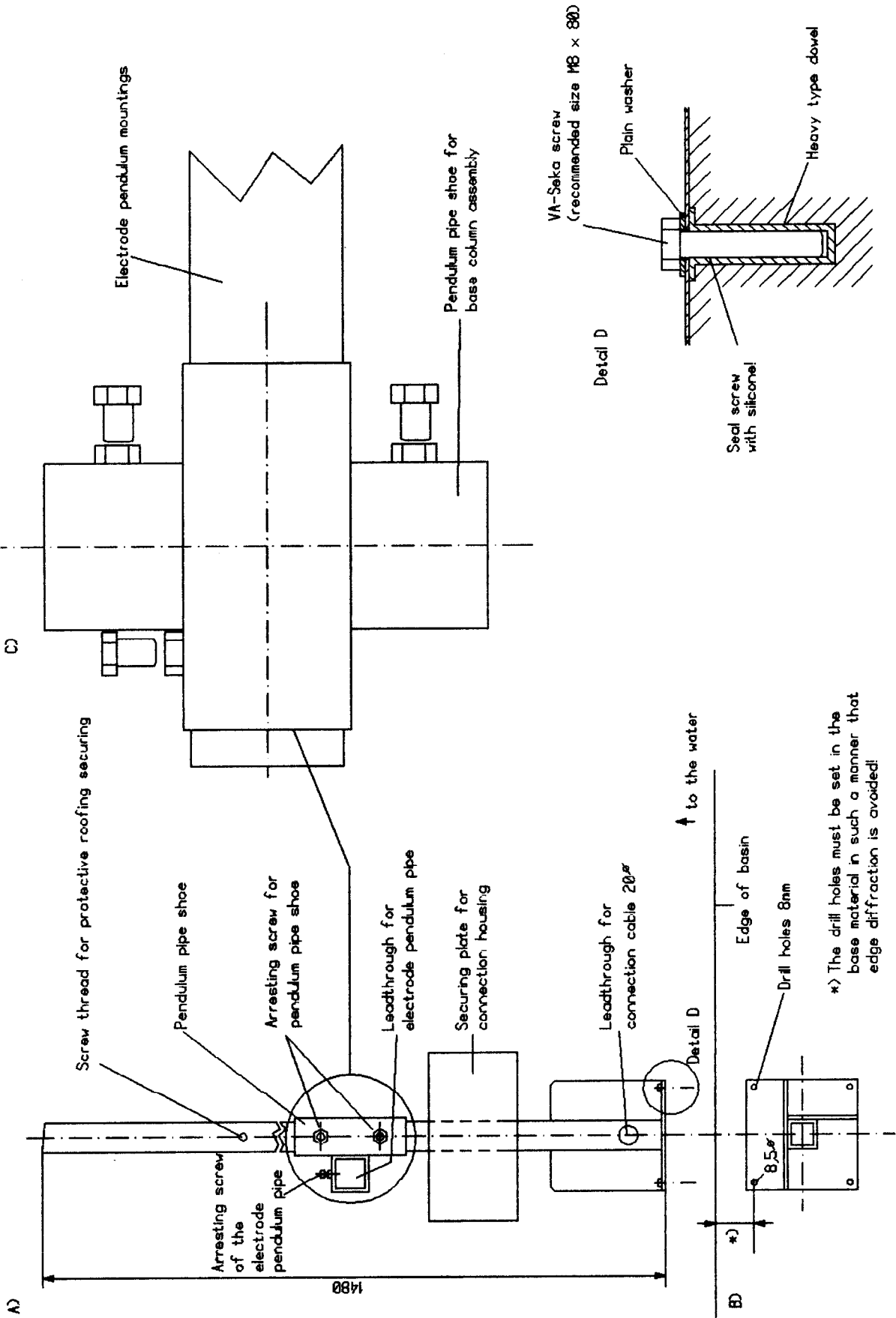


Figure 5: Notifications, causes and corrective measures

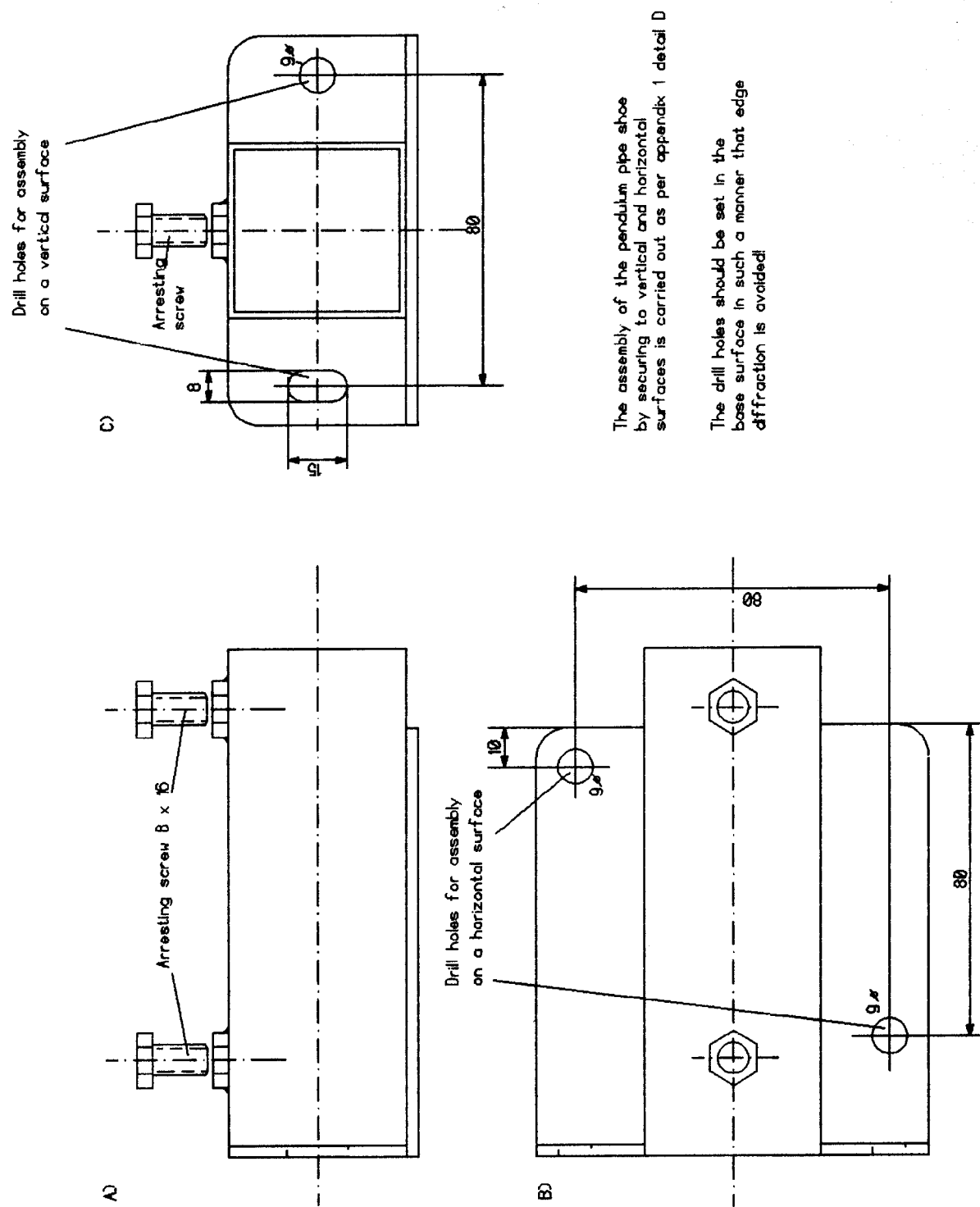
Should it not be possible to correct a disturbance, our service department is at your disposal on tel.: +49-881-62855 or facsimile +49-881-62815 or e-mail gimat2000@aol.com.

9 Appendix

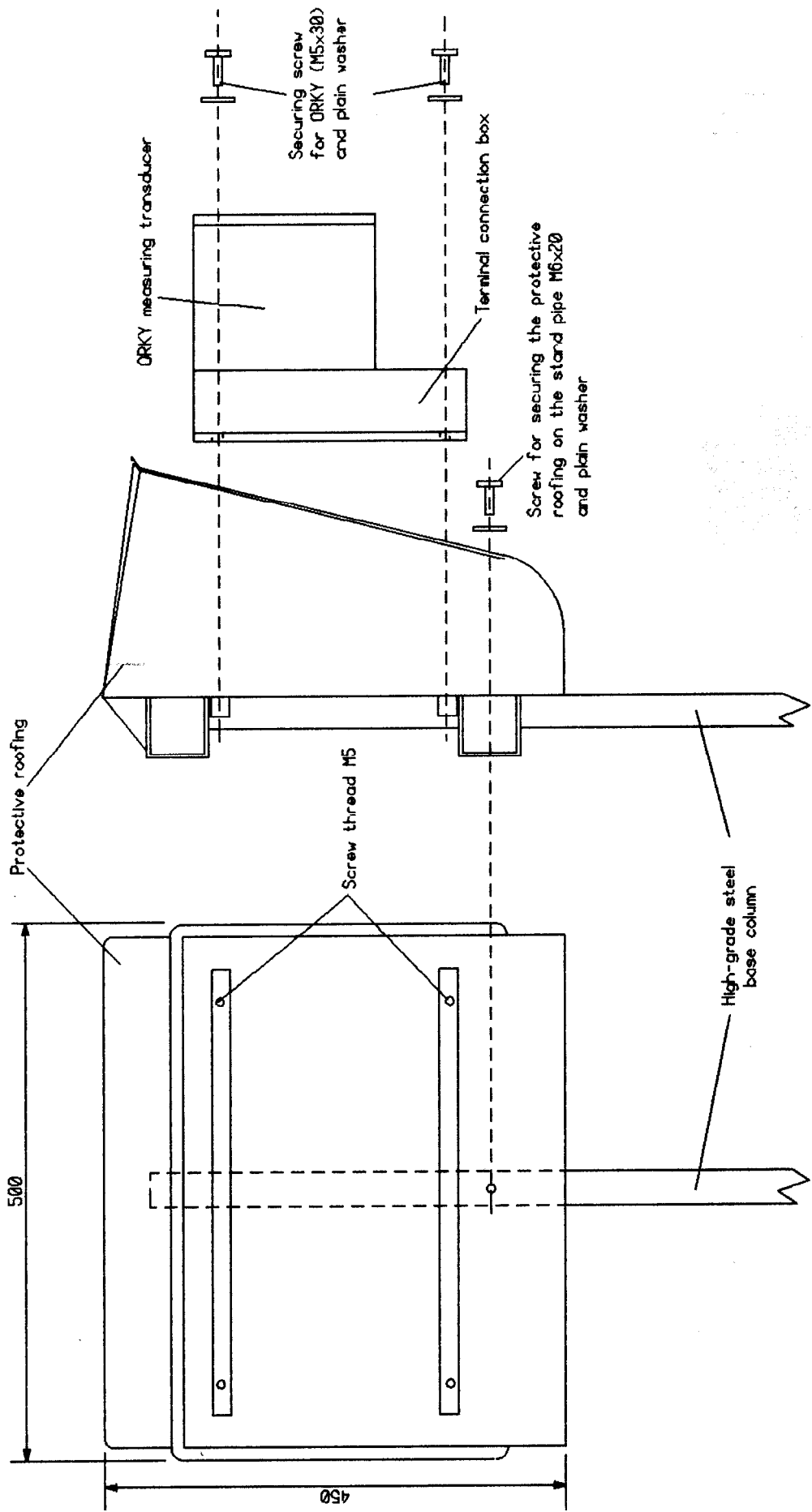
Appendix 1



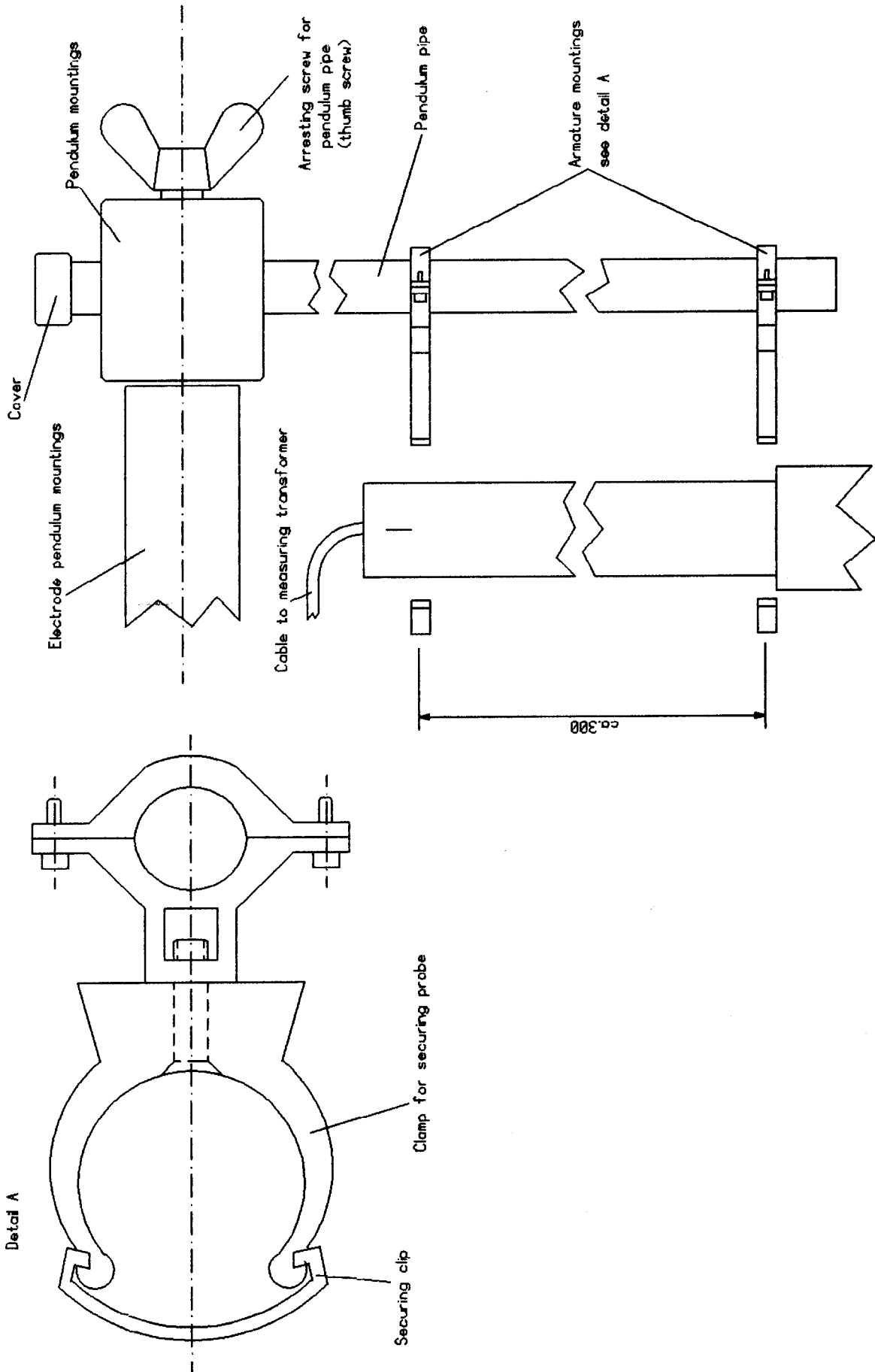
Appendix 2:



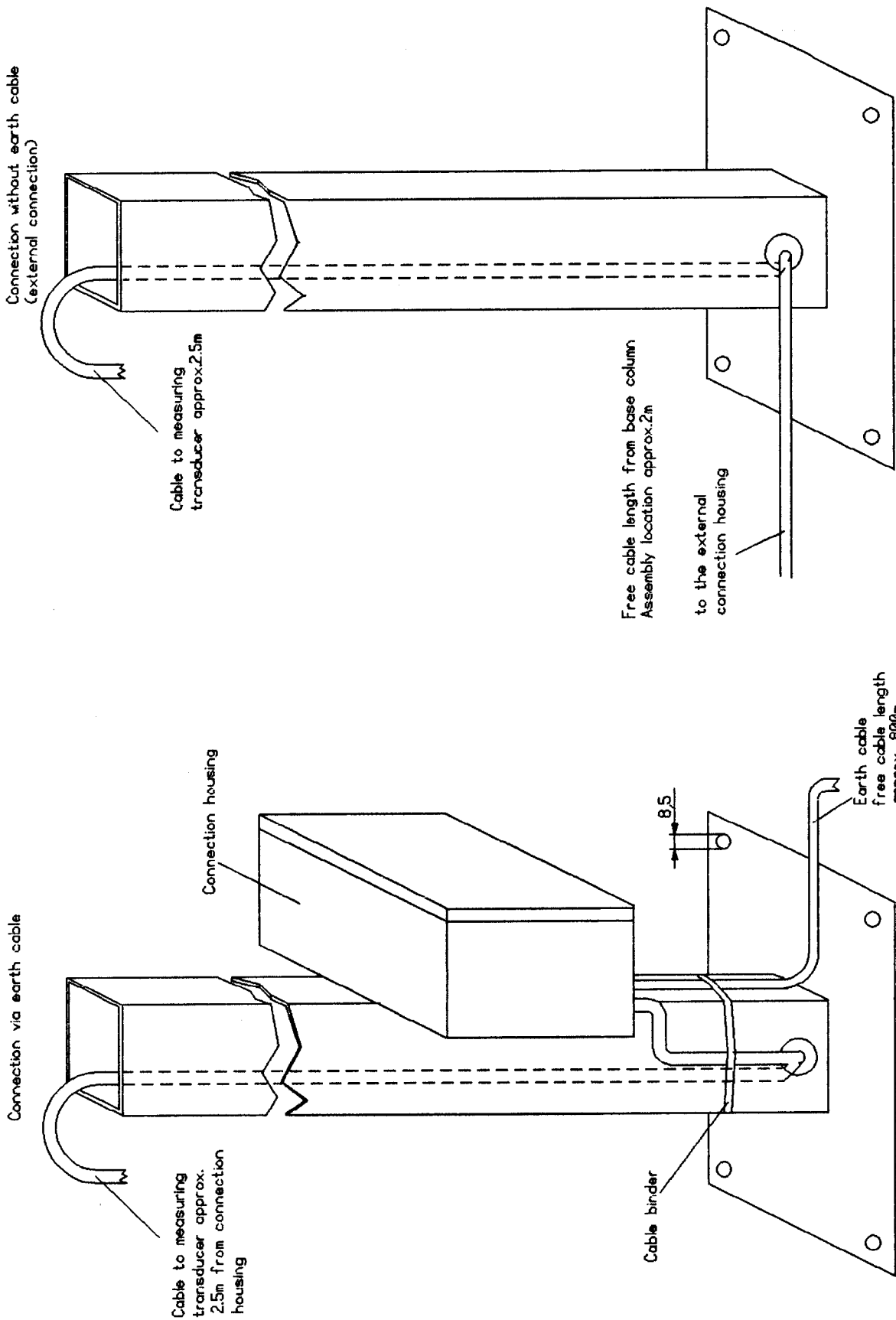
Appendix 3:



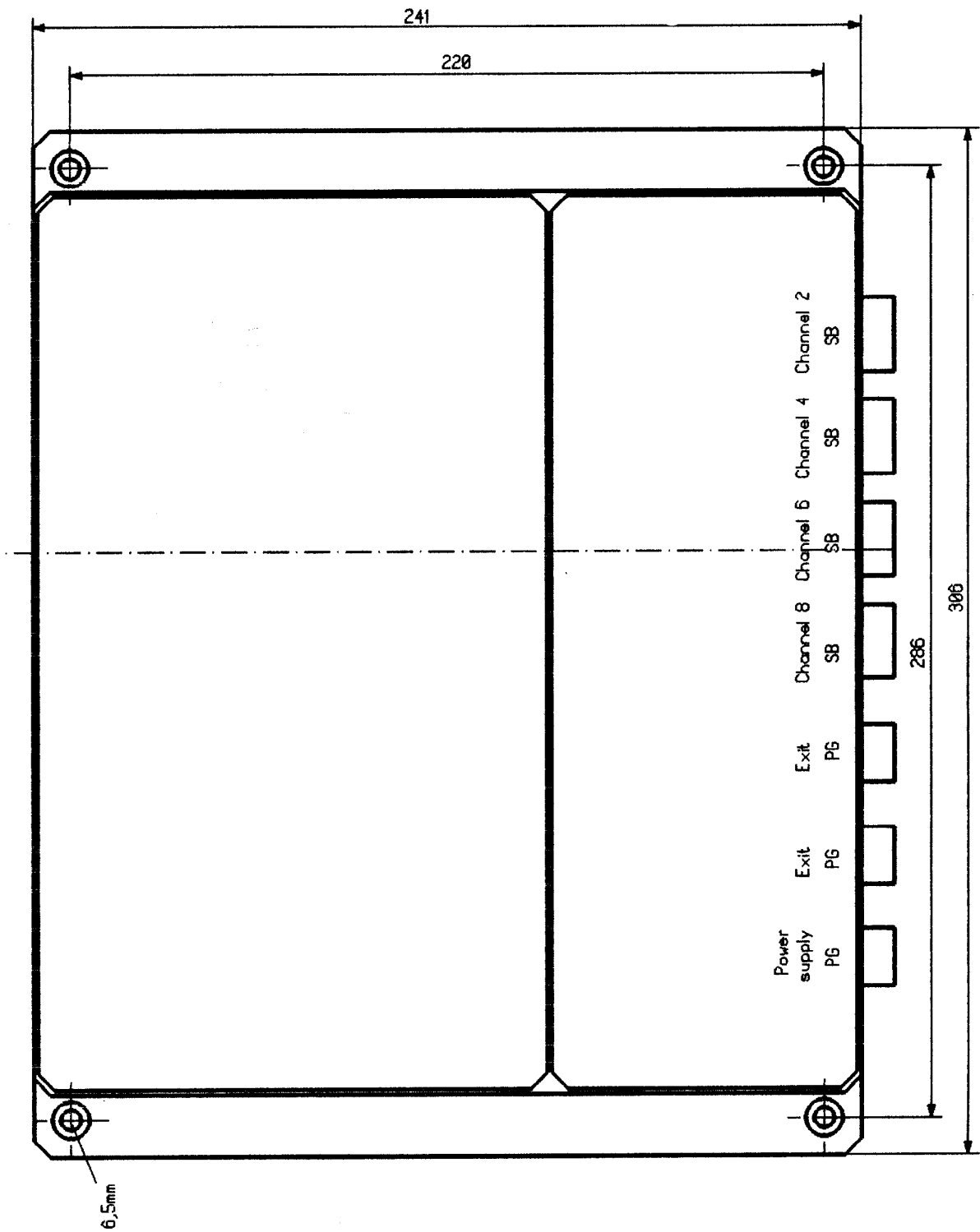
Appendix 4:



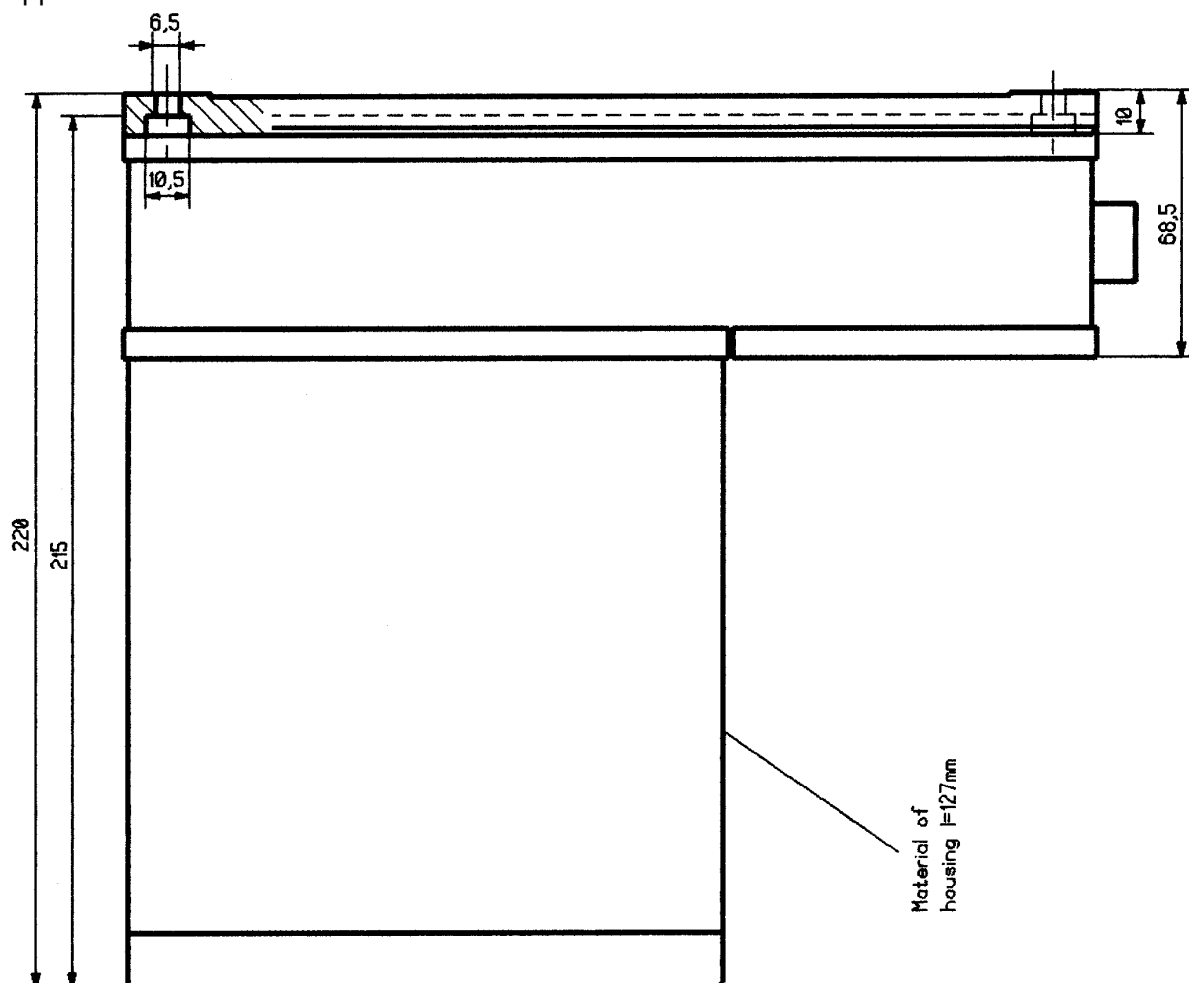
Appendix 5:



Appendix 6:

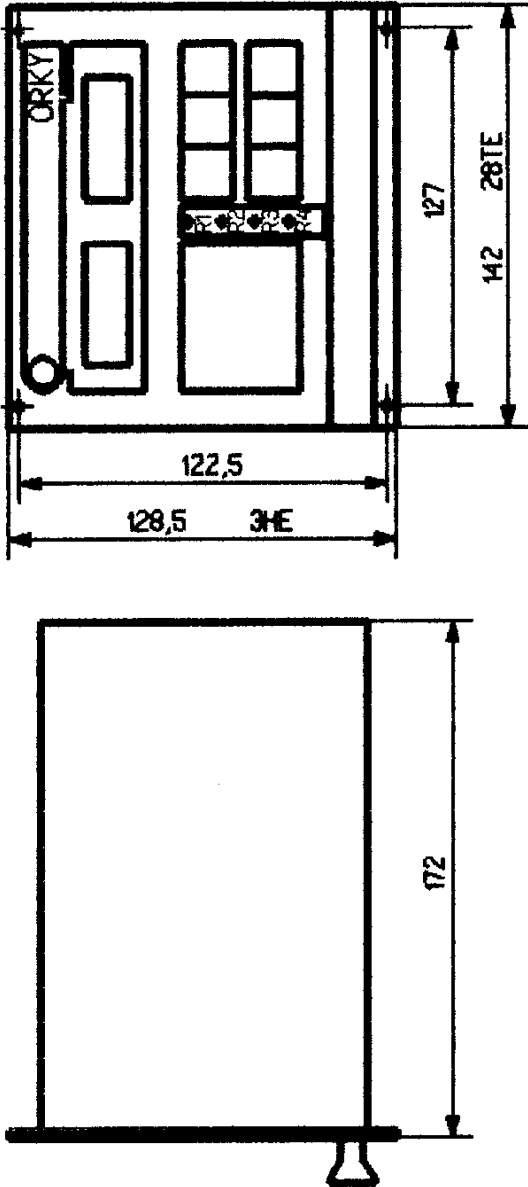


Appendix 7:



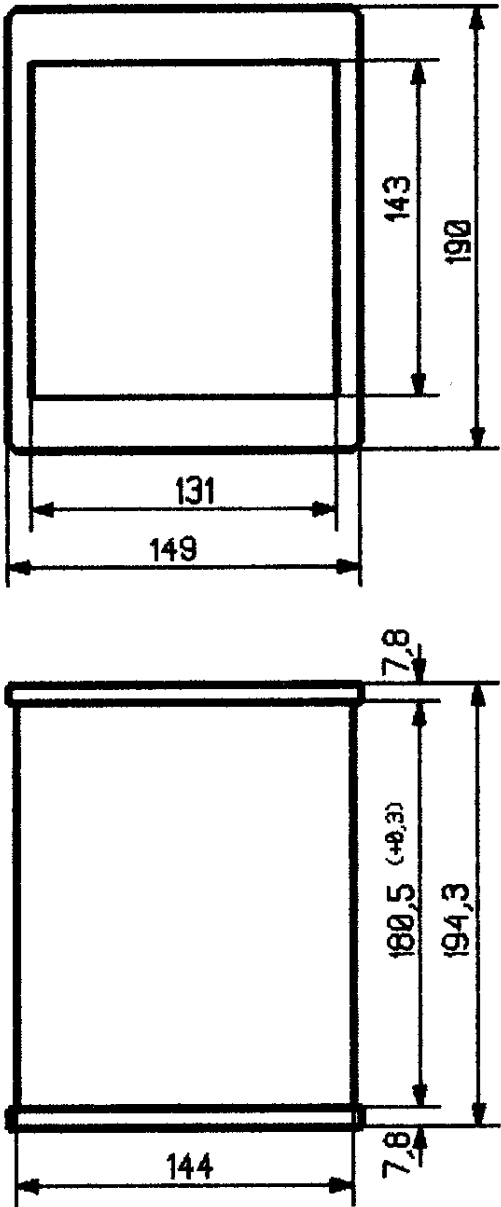
Appendix 8:

19" build in housing

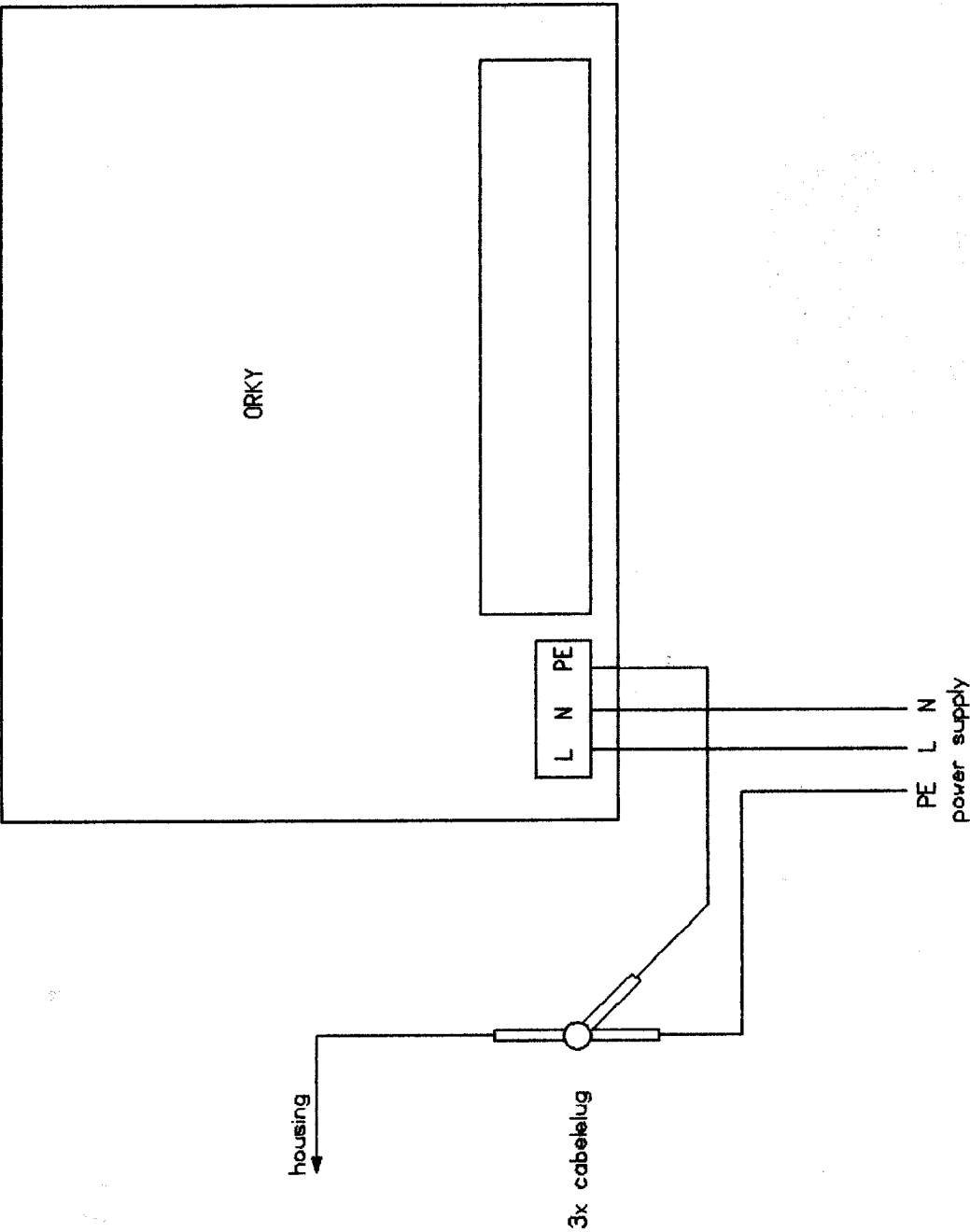


Appendix 9:

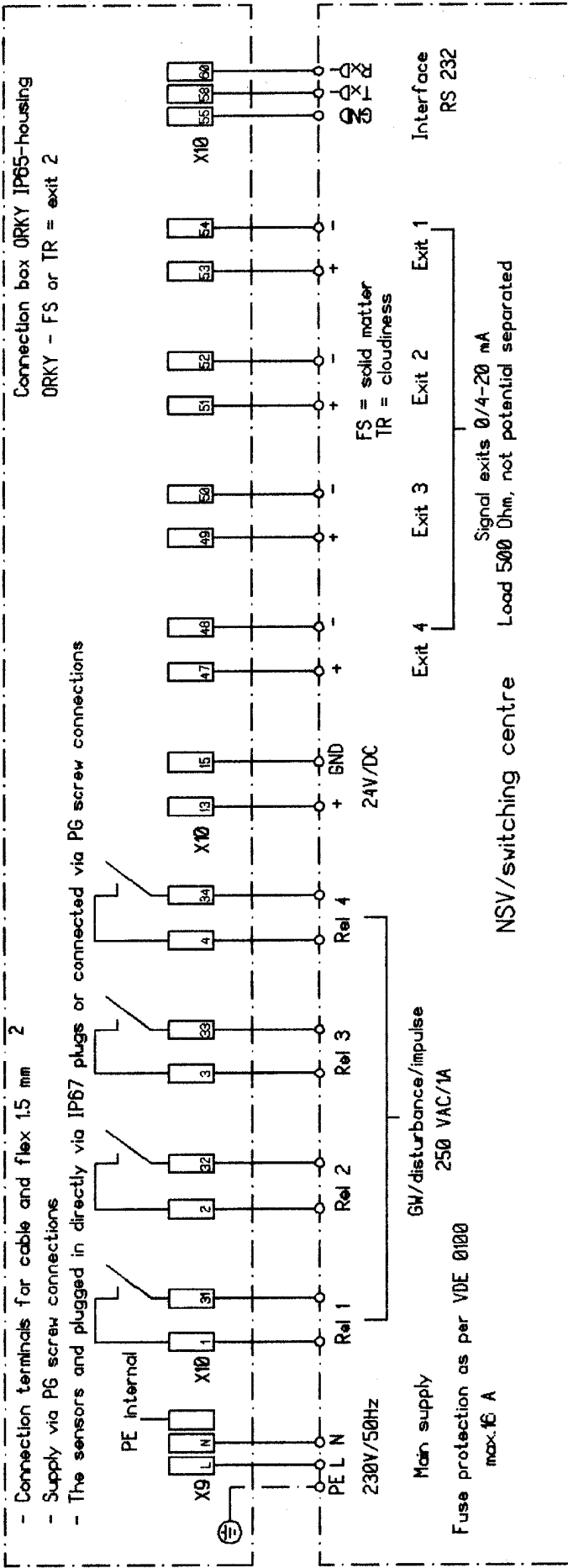
electrical control panel build in cabinet



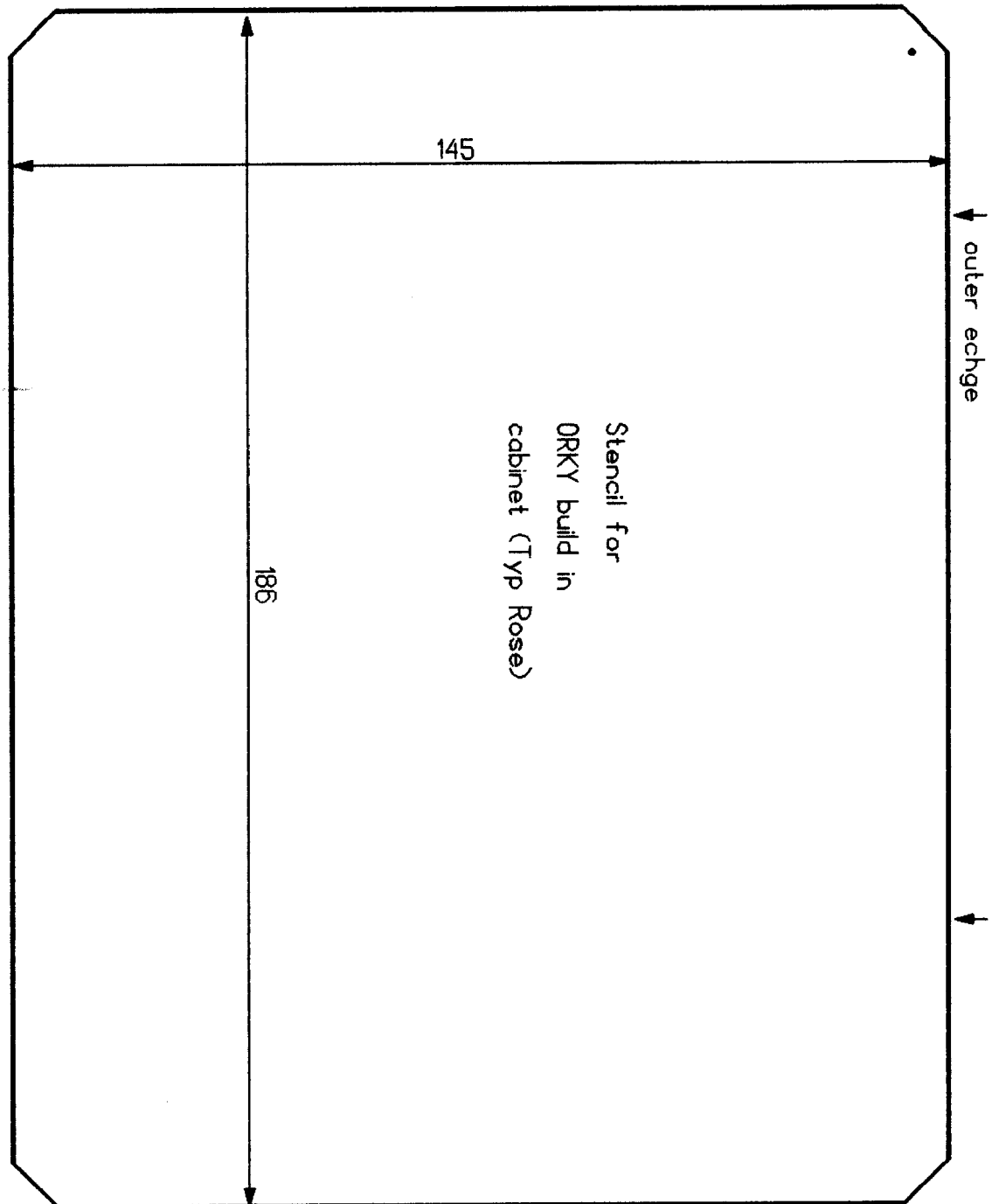
Appendix 10:



Appendix 11:



Appendix 12:



Appendix 13:

Assembling instruction for probe mountings in basins